

1-PHPStan

```
# phpstan.neon
parameters:
  level: 5
  paths:
    - src
  excludePaths:
  analyseAndScan:
    - src/Kernel.php
```

Réclamation Avant et après Optimisation :

```
PS C:\Users\Lamis\Desktop\INTG_F\apis\unibank\F\unibank\F\unibank> vendor/bin/phpstan analyse src/Controller/Client/ReclamationController.php --level=5 --memory-limit=512M
>>
PHP Warning: Module "mysqli" is already loaded in Unknown on line 0

Warning: Module "mysqli" is already loaded in Unknown on line 0
Note: Using configuration file C:\Users\Lamis\Desktop\INTG_F\apis\unibank\F\unibank\F\unibank\phpstan.neon.
1/1 [=====] 100%

-----
Line   ReclamationController.php
-----
:177   Parameter #1 $utilisateur of method App\Entity\Reclamation::setUtilisateur() expects App\Entity\Utilisateur|null,
       argument.type
:194   Left side of && is always true.
       booleanAnd.leftAlwaysTrue
       ⚠ Because the type is coming from a PHPDoc, you can turn off this check by setting treatPhpDocTypesAsCertain: false in your phpstan.neon.
:237   Call to an undefined method Symfony\Component\Security\Core\User\UserInterface::getIdUtilisateur().
       method.notFound
:261   Call to an undefined method Symfony\Component\Security\Core\User\UserInterface::getIdUtilisateur().
       method.notFound
:287   Left side of && is always true.
       booleanAnd.leftAlwaysTrue
       ⚠ Because the type is coming from a PHPDoc, you can turn off this check by setting treatPhpDocTypesAsCertain: false in your phpstan.neon.
:347   Call to an undefined method Symfony\Component\Security\Core\User\UserInterface::getIdUtilisateur().
       method.notFound
:376   Call to an undefined method Symfony\Component\Security\Core\User\UserInterface::getIdUtilisateur().
       method.notFound
:406   Call to an undefined method App\Service\ImageModerationService::verifierContenuImage().
       method.notFound
-----

PS C:\Users\Lamis\Desktop\INTG_F\apis\unibank\F\unibank\F\unibank> vendor/bin/phpstan analyse src/Controller/Client/ReclamationController.php --level=5 --memory-limit=512M
>>
>>
PHP Warning: Module "mysqli" is already loaded in Unknown on line 0

Warning: Module "mysqli" is already loaded in Unknown on line 0
Note: Using configuration file C:\Users\Lamis\Desktop\INTG_F\apis\unibank\F\unibank\F\unibank\phpstan.neon.
1/1 [=====] 100%

[OK] No errors

PS C:\Users\Lamis\Desktop\INTG_F\apis\unibank\F\unibank\F\unibank>
```

```
PS C:\Users\Lamis\Desktop\INTG_F\apis\unibank\F\unibank\F\unibank+> vendor/bin/phpstan analyse src/Entity/Reclamation.php --level=5 --memory-limit=512M
>>
PHP Warning: Module "mysqli" is already loaded in Unknown on line 0

Warning: Module "mysqli" is already loaded in Unknown on line 0
Note: Using configuration file C:\Users\Lamis\Desktop\INTG_F\apis\unibank\F\unibank\F\unibank+\.phpstan.neon.
1/1 [=====] 100%

-----
Line    Reclamation.php
-----
:18     Property App\Entity\Reclamation::$id (int|null) is never assigned int so it can be removed from the property type.
       property.unusedType
-----

[ERROR] Found 1 error
```

```
PS C:\Users\Lamis\Desktop\INTG_F\apis\unibank\F\unibank\F\unibank+> vendor/bin/phpstan analyse src/Entity/Reclamation.php --level=5 --memory-limit=512M
>>
PHP Warning: Module "mysqli" is already loaded in Unknown on line 0

Warning: Module "mysqli" is already loaded in Unknown on line 0
Note: Using configuration file C:\Users\Lamis\Desktop\INTG_F\apis\unibank\F\unibank\F\unibank+\.phpstan.neon.
1/1 [=====] 100%

[OK] No errors
```

2 — Tests Unitaires

1/Gestion Réclamation :

- Règle métier 1 : Maximum 3 réclamations par jour
- Règle métier 2 : Modification impossible après 7 jours si en attente

```
PS C:\Users\Lamis\Desktop\INTG_F\apis\unibank\F\unibank\F\unibank+> php bin/console make:test TestCase ReclamationBusinessService
>>
PHP Warning: Module "mysqli" is already loaded in Unknown on line 0

Warning: Module "mysqli" is already loaded in Unknown on line 0
created: tests/ReclamationBusinessServiceTest.php

Success!

Next: Open your new test class and start customizing it.
Find the documentation at https://symfony.com/doc/current/testing.html#unit-tests
PS C:\Users\Lamis\Desktop\INTG_F\apis\unibank\F\unibank\F\unibank+>
```

```

PS C:\Users\Lamis\Desktop\INTG_F\apis\unibank\F\unibank\F\unibank+> vendor\bin\phpunit tests\Unit\Entity\ReclamationTest.php
PHP Warning: Module "mysqli" is already loaded in Unknown on line 0

Warning: Module "mysqli" is already loaded in Unknown on line 0
PHPUnit 11.5.55 by Sebastian Bergmann and contributors.

Runtime:      PHP 8.2.30
Configuration: C:\Users\Lamis\Desktop\INTG_F\apis\unibank\F\unibank\F\unibank+\phpunit.dist.xml

....

4 / 4 (100%)

Time: 00:00.047, Memory: 10.00 MB

OK (4 tests, 19 assertions)
PS C:\Users\Lamis\Desktop\INTG_F\apis\unibank\F\unibank\F\unibank+>

```

```

PS C:\Users\Lamis\Desktop\INTG_F\apis\unibank\F\unibank\F\unibank+> vendor\bin\phpunit --no-configuration --bootstrap vendor/autoload.php ReclamationTest.php
PHP Warning: Module "mysqli" is already loaded in Unknown on line 0

Warning: Module "mysqli" is already loaded in Unknown on line 0
PHPUnit 11.5.55 by Sebastian Bergmann and contributors.

Runtime:      PHP 8.2.30

.....

8 / 8 (100%)

Time: 00:00.086, Memory: 10.00 MB

OK (8 tests, 8 assertions)
PS C:\Users\Lamis\Desktop\INTG_F\apis\unibank\F\unibank\F\unibank+>

```

```

PS C:\Users\Lamis\Desktop\INTG_F\apis\unibank\F\unibank\F\unibank+> vendor\bin\phpunit --no-configuration --bootstrap vendor/autoload.php tests\Unit\Entity\ReponseTest.php
PHP Warning: Module "mysqli" is already loaded in Unknown on line 0

Warning: Module "mysqli" is already loaded in Unknown on line 0
PHPUnit 11.5.55 by Sebastian Bergmann and contributors.

Runtime:      PHP 8.2.30

.....

8 / 8 (100%)

Time: 00:00.038, Memory: 10.00 MB

OK (8 tests, 14 assertions)
PS C:\Users\Lamis\Desktop\INTG_F\apis\unibank\F\unibank\F\unibank+>

```

3-Problèmes détectés (Doctrine Doctor)

1/

Total Issues	Critical	Warnings	Info	Queries Analyzed	Profiler Overhead
39	3	20	16	1	3,784.10 ms

Performance (0)

Security (5)

Integrity (28)

Configuration (6)

Slowest Queries

INTEGRITY

Missing orphanRemoval on Composition Relationship

OneToMany field `reponses` in entity `App\Entity\Reclamation` appears to be a composition relationship but lacks `orphanRemoval=true`. This leaves orphaned records in the database when children are removed from the collection.

Hide suggested fix ^

Hide backtrace ^

Missing orphanRemoval

Removing from collection leaves orphan records. Add `orphanRemoval=true` for composition relationships.

Le problème est disparu :

Performance (0)

Security (5)

Integrity (27)

Configuration (6)

Slowest Queries

INTEGRITY

Foreign Key Missing _id Suffix 11

Foreign key 'id_utilisateur' should end with '_id' suffix according to Doctrine conventions.

Suggested fix ^

INTEGRITY

ORM Cascade / Database onDelete Mismatch

Field `reponses` in `App\Entity\Reclamation` has `cascade="remove"` in ORM but no onDelete constraint in database. ORM deletes work, but direct SQL DELETES may cause FK constraint violations.

Suggested fix ^

2/

Performance (0) Security (5) Integrity (27) Configuration (6) Slowest Queries

INTEGRITY Foreign Key Missing _id Suffix ? 11

Foreign key 'id_utilisateur' should end with '_id' suffix according to Doctrine conventions.

Suggested fix

INTEGRITY ORM Cascade / Database onDelete Mismatch

Field `reponses` in `App\Entity\Reclamation` has `cascade="remove"` in ORM but no onDelete constraint in database. ORM deletes work, but direct SQL DELETes may cause FK constraint violations.

Hide suggested fix

Fix ORM Cascade / Database onDelete Mismatch

ORM cascade `remove` conflicts with DB onDelete `NO ACTION`. Deletion behavior varies by method (ORM vs SQL).

Le problème est disparu :

Performance (0) Security (5) Integrity (26) Configuration (6) Slowest Queries

INTEGRITY Foreign Key Missing _id Suffix ? 11

Foreign key 'id_utilisateur' should end with '_id' suffix according to Doctrine conventions.

Suggested fix

PROPERTY_TYPE_MISMATCH Type Mismatch: Utilisateur::\$role

Property Utilisateur::\$role has type mismatch:
Expected: string

Gestion de credit

1-PHPStan

Cette erreur est apparue parce que j'ai saisi une instruction de configuration PHPStan dans le terminal PowerShell, alors qu'elle devait être ajoutée dans le fichier `phpstan.neon`.

```

1 parameters:
2     level: 5
3     paths:
4         - src
5         - tests
6     excludePaths:
7         analyse:
8             - src/Kernel.php
9

```

Gestion de crédit Avant et après Optimisation :

```

PS C:\Users\asala\Downloads\INTG_F\INTG_F\apis\unibank+F\unibank+F\unibank+> v
endor\bin\phpstan analyse src\Entity\Credit.php src\Controller\Client\CreditCo
ntroller.php tests\Unit\Entity\CreditTest.php --level=5
>>
Note: Using configuration file C:\Users\asala\Downloads\INTG_F\INTG_F\apis\uni
bank+F\unibank+F\unibank+\phpstan.neon.
 3/3 [=====] 100%

[OK] No errors

PS C:\Users\asala\Downloads\INTG_F\INTG_F\apis\unibank+F\unibank+F\unibank+>

```

PHPStan a signalé une remarque dans l'entité Credit concernant le type de la propriété idCredit, déclaré comme int|null alors qu'aucune affectation explicite de type int n'est détectée dans le code.

```

PS C:\Users\asala\Downloads\INTG_F\INTG_F\apis\unibank\F\unibank\F\unibank+> vendor\bin\phpunit --no-configuration tests\Unit\Entity\CreditTest.php
>>
PHPUnit 11.5.55 by Sebastian Bergmann and contributors.

Runtime:       PHP 8.2.12

.....                                         5 / 5 (100%)
Time: 00:00.013, Memory: 10.00 MB

OK (5 tests, 21 assertions)
PS C:\Users\asala\Downloads\INTG_F\INTG_F\apis\unibank\F\unibank\F\unibank+> vendor\bin\phpstan analyse src\Entity\Credit.php src\Controller\Client\CreditController.php tests\Unit\Entity\CreditTest.php --level=5
>>
Note: Using configuration file C:\Users\asala\Downloads\INTG_F\INTG_F\apis\unibank\F\unibank\F\unibank+\phpstan.neon.
 3/3 [=====] 100%

[OK] No errors
PS C:\Users\asala\Downloads\INTG_F\INTG_F\apis\unibank\F\unibank\F\unibank+>

```

Cette

commande a permis d'exécuter une analyse statique du code avec PHPStan sur l'entité Credit, le contrôleur client et le fichier de test. Le résultat affiché à 100 % indique que l'analyse a été terminée avec succès sur les trois fichiers ciblés.

2 — Tests Unitaires

1/Gestion Crédit :

- **Règle métier 1** : Calcul automatique de la mensualité et du montant total du crédit
- **Règle métier 2** : Le coût total du crédit est calculé à partir du montant total moins le montant emprunté
- **Règle métier 3** : Une demande de crédit possède par défaut le statut PENDING
- **Règle métier 4** : Si le taux d'intérêt est nul, la mensualité est calculée par division simple du montant sur la durée

```
PS C:\Users\asala\Downloads\INTG_F\INTG_F\apis\unibank+F\unibank+F\unibank+> p
hp bin/console make:test TestCase CreditTest
>>
created: tests/CreditTest.php

Success!

Next: Open your new test class and start customizing it.
Find the documentation at https://symfony.com/doc/current/testing.html#unit-tests
PS C:\Users\asala\Downloads\INTG_F\INTG_F\apis\unibank+F\unibank+F\unibank+>
```

```
endor\bin\phpunit --no-configuration tests\Unit\Entity\CreditTest.php
>>
PHPUnit 11.5.55 by Sebastian Bergmann and contributors.

Runtime:       PHP 8.2.12

.....                                     5 / 5 (100%)

Time: 00:00.013, Memory: 10.00 MB

OK (5 tests, 21 assertions)
PS C:\Users\asala\Downloads\INTG_F\INTG_F\apis\unibank+F\unibank+F\unibank+>
```



```

endor\bin\phpunit --no-configuration --bootstrap vendor/autoload.php tests\Unit\Entity\CreditTest.php
>>
PHPUnit 11.5.55 by Sebastian Bergmann and contributors.

Runtime:       PHP 8.2.12

.....                                         5 / 5 (100%)
Time: 00:00.014, Memory: 10.00 MB

OK (5 tests, 21 assertions)
PS C:\Users\asala\Downloads\INTG_F\INTG_F\apis\unibank+F\unibank+F\unibank+> vendor\bin\phpstan analyse src\Entity\Credit.php src\Controller\Client\CreditController.php tests\Unit\Entity\CreditTest.php --level=5
>>
Note: Using configuration file C:\Users\asala\Downloads\INTG_F\INTG_F\apis\unibank+F\unibank+F\unibank+\phpstan.neon.
 3/3 [=====] 100%

[OK] No errors

PS C:\Users\asala\Downloads\INTG_F\INTG_F\apis\unibank+F\unibank+F\unibank+>

```

La première commande vérifie la qualité du code avec PHPStan, tandis que la deuxième exécute les tests unitaires de l'entité Credit avec PHPUnit.

3-Problèmes détectés (Doctrine Doctor)

Last 10LatestSearch

Request / Response

Performance

Validator

Forms

Exception

Logs

Events

Routing

Cache

Translation

Security

Twig

HTTP Client

Doctrine

Doctrine Doctor
36

Debug

Messages

Doctrine Doctor

Database Engine

mysql

Database Version

10.4.32-MariaDB

Doctrine ORM Version

3.6.3

Total Issues

36

Critical

2

Warnings

16

Info

18

Queries Analyzed

0

Profiler Overhead

806.30 ms

Performance (0)

Security (5)

Integrity (23)

Configuration (8)

Slowest Queries

SECURITY

Unprotected sensitive field: CarteSecurity::\$cvv

The sensitive field "\$cvv" in entity "CarteSecurity" lacks serialization protection. Without @JsonIgnore or #[Ignore] annotations, this field will be included in JSON serialization, API responses, and logs. Additionally, consider using #[SensitiveParameter] on setter methods to prevent values from appearing in stack traces. Add appropriate annotations to prevent data leakage.

Suggested fix

Backtrace

SECURITY

Unprotected sensitive field: Reclamation::\$token

The sensitive field "\$token" in entity "Reclamation" lacks serialization protection. Without @JsonIgnore or #[Ignore] annotations, this field will be included in JSON serialization, API responses, and logs. Additionally, consider using #[SensitiveParameter] on setter methods to prevent values from appearing in stack traces. Add appropriate annotations to prevent data leakage.

Last 10

Latest

Search

Request / Response

Performance

Validator

Forms

Exception

Logs

Events

Routing

Cache

Translation

Security

Twig

HTTP Client

Doctrine

Entity Credit::\$typeContrat has type hint "App\Enum\TypeContrat" but Doctrine type "string" expects "string". This causes the UnitOfWork to detect false changes and execute unnecessary UPDATE statements on every flush.

Suggested fix

Backtrace

Entity Credit::\$statut has type hint "App\Enum\CreditStatus" but Doctrine type "string" expects "string". This causes the UnitOfWork to detect false changes and execute unnecessary UPDATE statements on every flush.

Suggested fix

Backtrace

Entity Credit::\$typeContrat has type hint "App\Enum\TypeContrat" but Doctrine type "string" expects "string". This causes the UnitOfWork to detect false changes and execute unnecessary UPDATE statements on every flush.

Suggested fix

Backtrace

mysql10.4.32-MariaDB3.6.3

Total Issues30

Critical2

Warnings10

Info18

Queries Analyzed0

Profiler Overhead463.50 ms

Performance (0)

Security (5)

Integrity (17)

Configuration (8)

Slowest Queries

INTEGRITY Foreign Key Missing _id Suffix

Foreign key 'id_utilisateur' should end with '_id' suffix according to Doctrine conventions.

Suggested fix

INTEGRITY Nullable Creation Timestamp: CarteSecurity::\$createdAt

Field CarteSecurity::\$createdAt (creation timestamp) is nullable. Creation timestamp should never be null - every entity has a creation time. Mark this field as NOT NULL and initialize it in the constructor or use lifecycle callbacks.

Suggested fixBacktrace

INTEGRITY Missing Blameable Trait: CarteSecurity

Entity CarteSecurity has timestamp field(s) (createdAt) but no blameable field(s) (createdBy/updatedAtBy). Consider using

Last 10LatestSearch

Request / Response

Performance

Validator

Forms

Exception

Logs11

Events

Routing

Cache

Translation

Security

Twig

HTTP Client

Doctrine

Doctrine Doctor30

Debug

Messages

Taper ici pour rechercher

12:2129/04/2026